

Explainable Vision-Language Diagnostic Assistant for Chest Radiographs

An offline, edge-optimized medical AI solution empowering rural health workers with automated chest X-ray interpretation, visual heatmap reasoning, and priority triage scoring.

Group number: 4 Cluster - 3 Section – 7

Under the Guidance of:
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Problem Statement

BACKGROUND & CHALLENGE



Acute Radiologist Shortage: Primary health centers in rural regions lack specialist radiologists to read X-rays and detect conditions like pneumonia or tuberculosis.



Diagnostic Delays & Travel Burden: Patients travel vast distances to urban hospitals, delaying time-critical care and incurring heavy financial expense.



Black-Box AI Mistrust: Traditional AI systems provide class outputs without explanations, making local nurses reluctant to rely on machine predictions.





1. Multimodal Diagnosis

To develop a vision-language assistant to generate plain-language diagnostic descriptions (e.g., "early signs of pneumonia") alongside class confidence scores.



2. Visual Explanation

To incorporate Grad-CAM heatmaps and attention overlays that circle the exact anatomical regions of concern, giving healthcare workers verifiable proof.



3. Automated Triage & Priority Guidance

To compute case urgency flags (e.g., "Please refer to specialist within 24 hours") so rural health workers can prioritize critical emergency cases immediately.

Literature Survey & Research Gap

Research Work / Model	Key Contribution	Limitation / Research Gap	Research Paper
ResNet & DenseNet	Strong CNN-based image classification and feature extraction.	Mainly focused on classification; limited report generation and multimodal reasoning.	ResNet Paper / DenseNet Paper
CheXNet	Detects pneumonia and other thoracic diseases from chest X-rays using deep CNNs.	Primarily classification-based and lacks comprehensive multimodal reasoning.	CheXNet Paper
Grad-CAM	Generates heatmaps showing image regions responsible for model predictions.	Visual explanation is not inherently connected to textual diagnosis or clinical reasoning.	Grad-CAM Paper
BLIP-2	Connects image encoders with large language models for vision-language understanding and generation.	Large language models can still require substantial computational resources.	BLIP-2 Paper
MedKLIP	Incorporates medical knowledge and aligns medical entities with visual image regions for X-ray diagnosis.	More complex architecture and computational requirements make lightweight deployment difficult.	MedKLIP Paper

Literature Survey & Research Gap

Research Work / Model	Key Contribution	Limitation / Research Gap	Research Paper
GLoRIA	Learns global and local image-text representations by associating image regions with words from radiology reports.	Requires paired image-report data and does not directly provide actionable triage recommendations.	GLoRIA Paper
ConVIRT	Uses paired medical images and text for contrastive learning and improves medical image representation with limited labels.	Focuses mainly on representation learning rather than complete diagnosis, explanation and triage.	ConVIRT Paper
MedCLIP	Uses medical knowledge-based contrastive learning with unpaired medical images and text.	Primarily supports representation learning and retrieval/classification rather than report generation and triage.	MedCLIP Paper
ChestX-ray8 / ChestX-ray14	Provides large-scale chest X-ray datasets for disease classification and weakly supervised localization.	Labels are largely derived from reports and the datasets alone do not provide multimodal clinical reasoning.	ChestX-ray8 Paper
Standalone Saliency Maps	Highlight important regions of medical images to support visual interpretation.	Usually lack textual diagnostic explanation, severity assessment and actionable triage guidance.	Grad-CAM Reference Paper

Innovation, Creativity & Novelty



Dual-Stream VLM Core

Combines Vision Transformers (ViT) for spatial feature extraction, CLIP for vision-text alignment, and BLIP-2 for diagnostic report synthesis into a unified framework.



Offline Edge Deployment

Pioneers 8-bit quantization on multimodal medical networks, enabling complete offline execution on mid-range Android mobile devices in zero-connectivity posts.



Transparent Reasoning

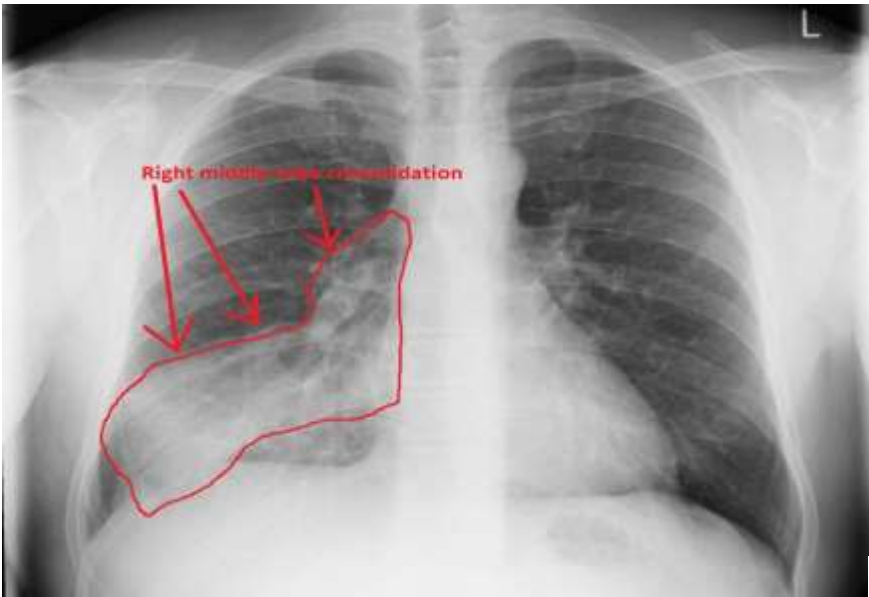
Generates dual Grad-CAM heatmaps and cross-attention overlays on the X-ray scan, bridging the trust gap between automated AI outputs and rural clinical workers.

Training Datasets

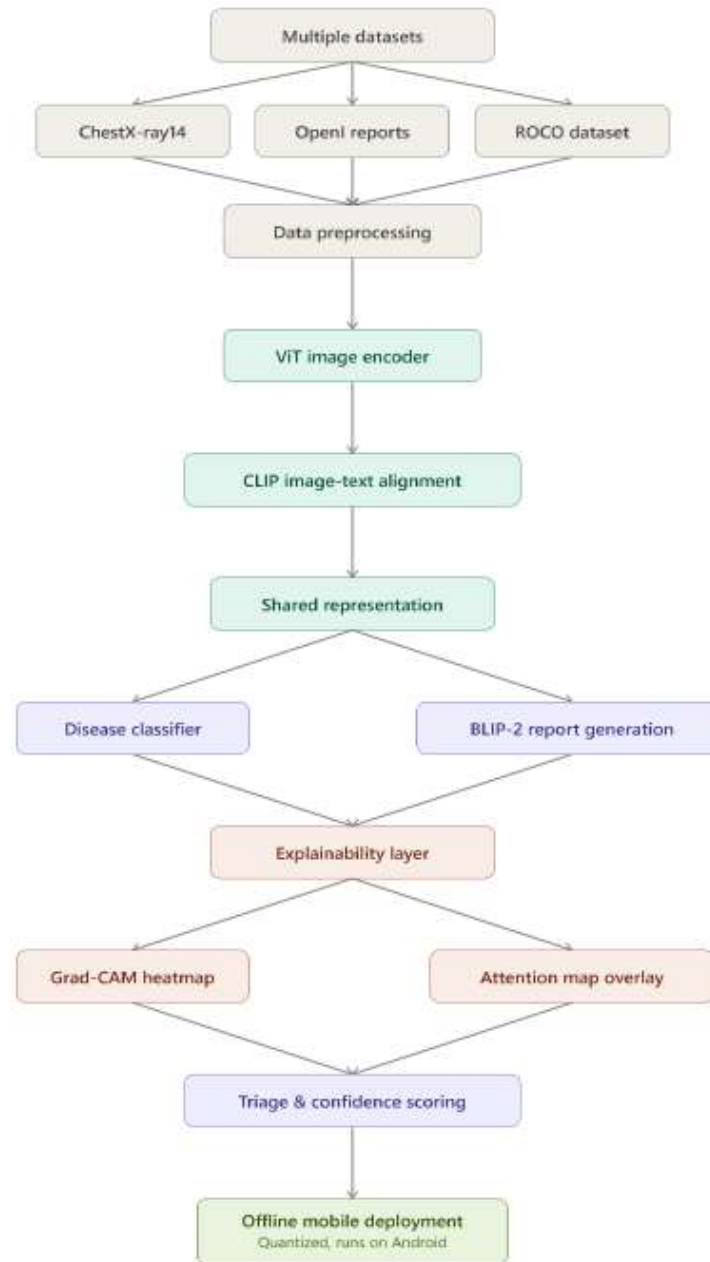
Dataset	Resource URL	Key Details
ChestX-ray14	NIH CC Box	112,120 chest X-rays with 14 disease labels.
OpenI (OpenAccess)	NLM OpenI	7,470 X-rays paired with clinical radiology reports.
ROCO Medical	GitHub / ROCO	Multimodal radiology dataset for text-image alignment.

Pipeline Flow

-  **Multi-Source Preprocessing:** Normalization & augmentations.
-  **ViT + CLIP Alignment:** Encodes spatial features into shared space.
-  **Dual Heads:** Disease Classifier + BLIP-2 Natural Text Generator.
-  **Explainability Layer:** Grad-CAM heatmaps & attention maps.
-  **Android Runtime:** Quantized model providing offline triage.



Flowchart



Project Plan & Evaluation Strategy

ROADMAP & METRICS

PHASE 1

Data Prep & Alignment

- Dataset curation (NIH, OpenI, ROCO)
- Data preprocessing & normalization
- CLIP vision-text alignment tuning

PHASE 2

Model & Explainability

- ViT encoder & BLIP-2 report integration
- Grad-CAM heatmap layer development
- Attention overlay generation

PHASE 3

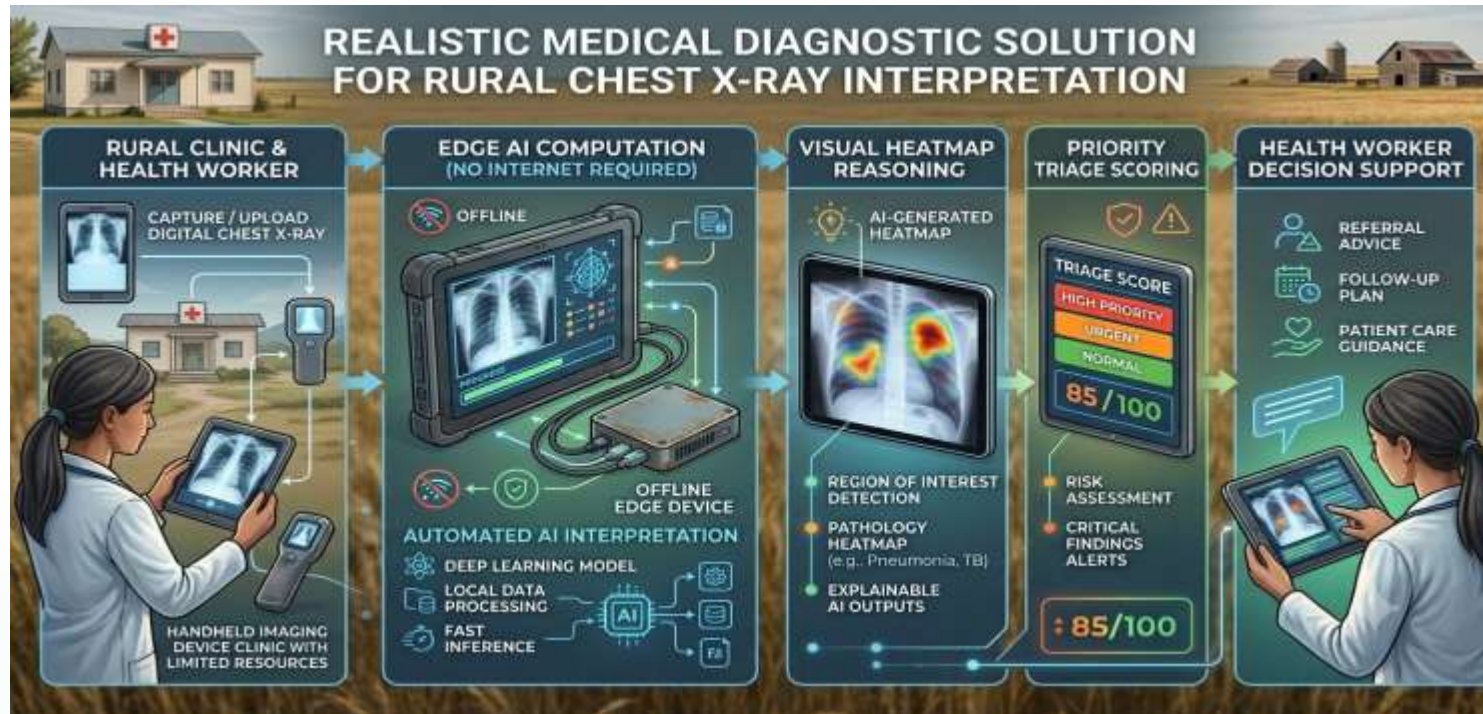
Optimization & Mobile

- INT8 model quantization
- ONNX / Android runtime integration
- Offline mobile performance testing

PHASE 4

Evaluation & Presentation

- Metrics: Accuracy, Precision, Recall, F1
- Heatmap localization verification



THANK YOU!

Our project aims to bridge the rural diagnostic gap by delivering an honest, explainable, and fully offline AI assistant — ensuring patients receive timely and safe care regardless of geographic limitations.